

Practical 7

System has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:

```
MINGW64/c/Users/STUDENTS/Desktop
Hit Percentage = 16.67%
Fault Percentage = 83.33%

STUDENTS@M309-63 MINGW64 ~/Desktop
$ cat lru.c
#include <stdio.h>

int main() {
    int frames, n, i, j, pos, faults = 0, hits = 0;

    printf("Enter number of frames: ");
    scanf("%d", &frames);

    printf("Enter number of page references: ");
    scanf("%d", &n);

    int pages[n];
    printf("Enter the page reference string:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    int frame[frames], time[frames];

    for(i = 0; i < frames; i++) {
        frame[i] = -1;
        time[i] = 0;
    }

    int counter = 0;

    for(i = 0; i < n; i++) {
        int found = 0;

        for(j = 0; j < frames; j++) {
            if(frame[j] == pages[i]) {
                hits++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

        if(!found) {
            faults++;

            int min = time[0];
            pos = 0;

            for(j = 1; j < frames; j++) {
                if(time[j] < min) {
                    min = time[j];
                    pos = j;
                }
            }

            frame[pos] = pages[i];
            counter++;
            time[pos] = counter;
        }
    }

    printf("\nTotal Page Hits = %d", hits);
    printf("\nTotal Page Faults = %d", faults);

    float hit_ratio = (float)hits / n * 100;
    float fault_ratio = (float)faults / n * 100;

    printf("\nHit Percentage = %.2f%%", hit_ratio);
    printf("\nFault Percentage = %.2f%%\n", fault_ratio);

    return 0;
}
```

Output:

```
MINGW64/c/Users/STUDENTS/Desktop
STUDENTS@M309-63 MINGW64 ~
$ gcc --version
gcc.exe (MinGW.org GCC-6.3.0-1) 6.3.0
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warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

STUDENTS@M309-63 MINGW64 ~
$
STUDENTS@M309-63 MINGW64 ~
$ cd ~/Desktop
STUDENTS@M309-63 MINGW64 ~/Desktop
$ nano lru.c
STUDENTS@M309-63 MINGW64 ~/Desktop
$ gcc lru.c -o lru
Enter number of frames: 3
Enter number of page references: 12
Enter the page reference string:
1 2 3 4 1 2 5 1 2 3 4 5

Total Page Hits = 2
Total Page Faults = 10
Hit Percentage = 16.67%
Fault Percentage = 83.33%

STUDENTS@M309-63 MINGW64 ~/Desktop
$
```